

END-TO-END DATA ANALYTICS PROJECT

Ecommerce Analytics

From SQL Server to Power BI to Python EDA



SQL Server



Power BI



Python EDA



ECOMMERCE ANALYTICS DASHBOARD

PROJECT OVERVIEW



1. PROJECT OBJECTIVE

To analyze ecommerce sales, customer behavior, product performance, payment trends, and business KPIs through an interactive Power BI dashboard that helps management make data-driven decisions.



2. BUSINESS QUESTIONS

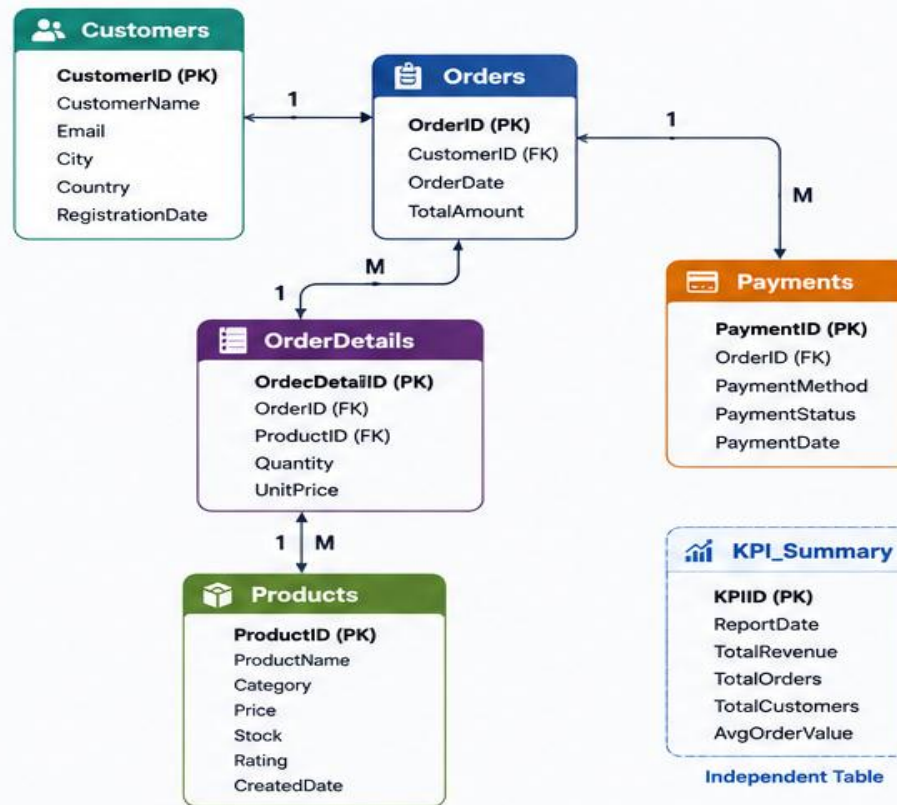
- What is the total revenue and number of orders?
- How many customers are registered?
- Which product category and products perform best?
- Which city has the most customers?
- Which payment method is used most?
- What is the payment success rate?
- What is the average order value?
- How is revenue changing over time?



3. DATA SOURCES

- **Products** – Product master data
- **Customers** – Customer information
- **Orders** – Order headers
- **OrderDetails** – Order line items
- **Payments** – Payment transactions
- **KPI_Summary** – Monthly KPI snapshot

4. DATA MODEL (RELATIONSHIPS)



5. KEY METRICS (KPIs)



6. DASHBOARD PAGES



Overview (Executive Dashboard)

High-level KPIs, revenue trend, category breakdown, top products, payment status.



Sales Analysis

Sales performance by product, category, monthly trends and top products.



Customer & Product Analysis

Customer insights, city/country analysis, product performance.



Payment Analytics

Payment method distribution, payment status and success rate.



7. INTERACTIVE FEATURES

- ✓ Slicers: Date, Category, Product, City, Country, Payment Method, Payment Status, Customer
- ✓ Drill Through & Tooltips
- ✓ Bookmarks & Buttons for Navigation
- ✓ Filters (Report, Page, Visual Level)
- ✓ Conditional Formatting & KPIs



8. TOOLS & TECHNOLOGIES



Power BI

Data Visualization & Dashboarding



SQL Server

Data Source & Data Modeling



VALUE DELIVERED

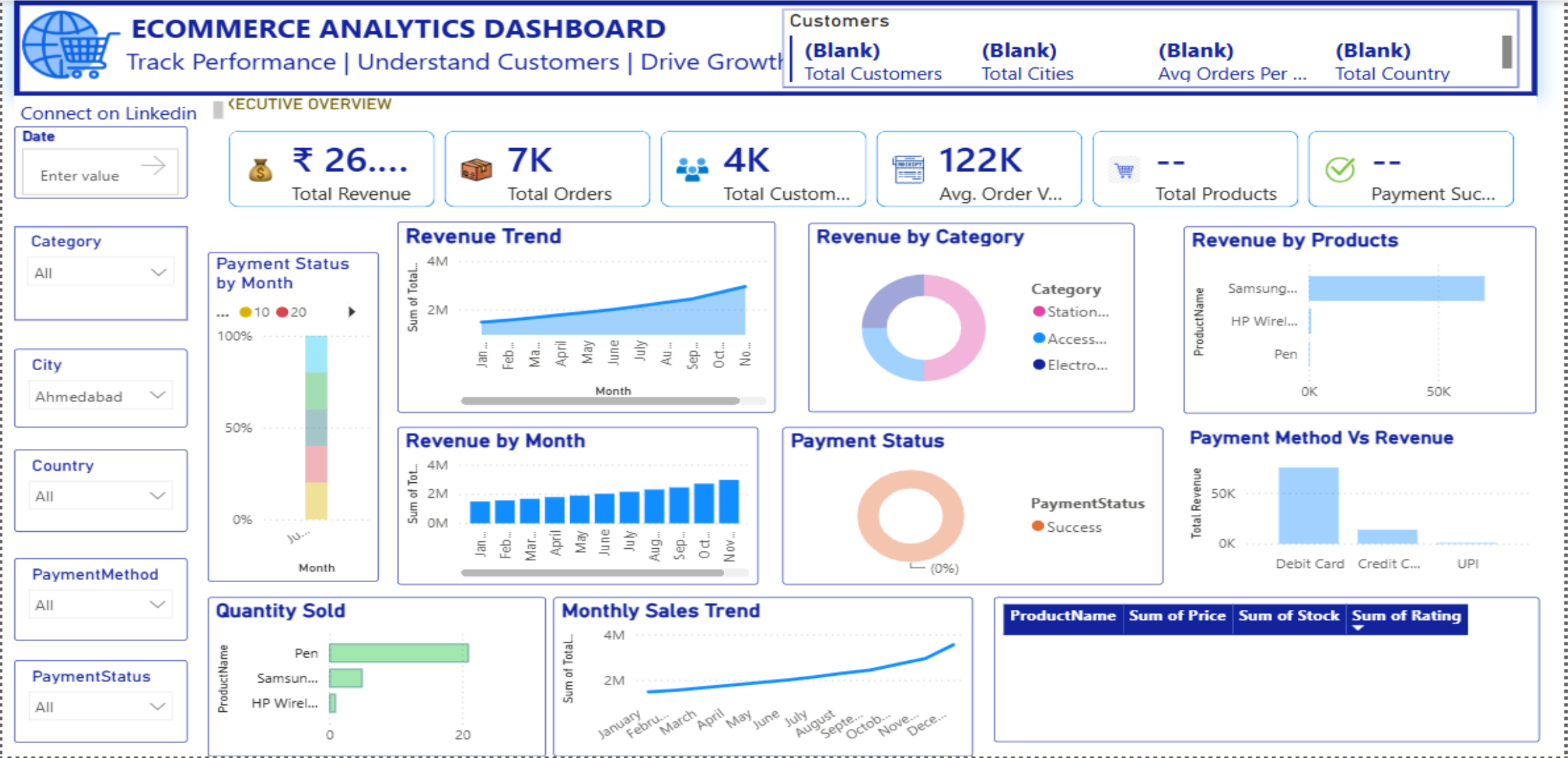
This dashboard provides end-to-end visibility of the ecommerce business, helping stakeholders monitor performance, identify opportunities, track KPIs, and make informed, data-driven decisions.

Data Pipeline



Each stage hands off a clean, structured artifact to the next — a relational schema feeds the BI model, which mirrors the same KPIs validated independently in the Python analysis.

Power BI Dashboard



Cross-Filtering with Slicers

Four slicers let a viewer drill into any segment of the business without writing a single query.



Order Date

Date-range slider across the full June 1–15 window



Payment Method

Credit Card, UPI, Debit Card, Net Banking, COD



Payment Status

Success, Pending, Failed



Product Category

Electronics, Accessories, Wearables, Smart Devices, Storage

Sync Slicers (View tab) keeps Order Date and Payment filters consistent across both report pages.

Data Cleaning in Pandas

The raw exports were validated and cleaned before any chart was built — every fix below was driven by an actual issue found in the data, not a generic checklist.



Clean keys

30 orders match 30 payments 1-to-1 on OrderID — zero missing or duplicate keys.



Duplicate catalog rows

39 raw product rows collapsed to 20 distinct products after removing duplicate reloads.



Inconsistent record

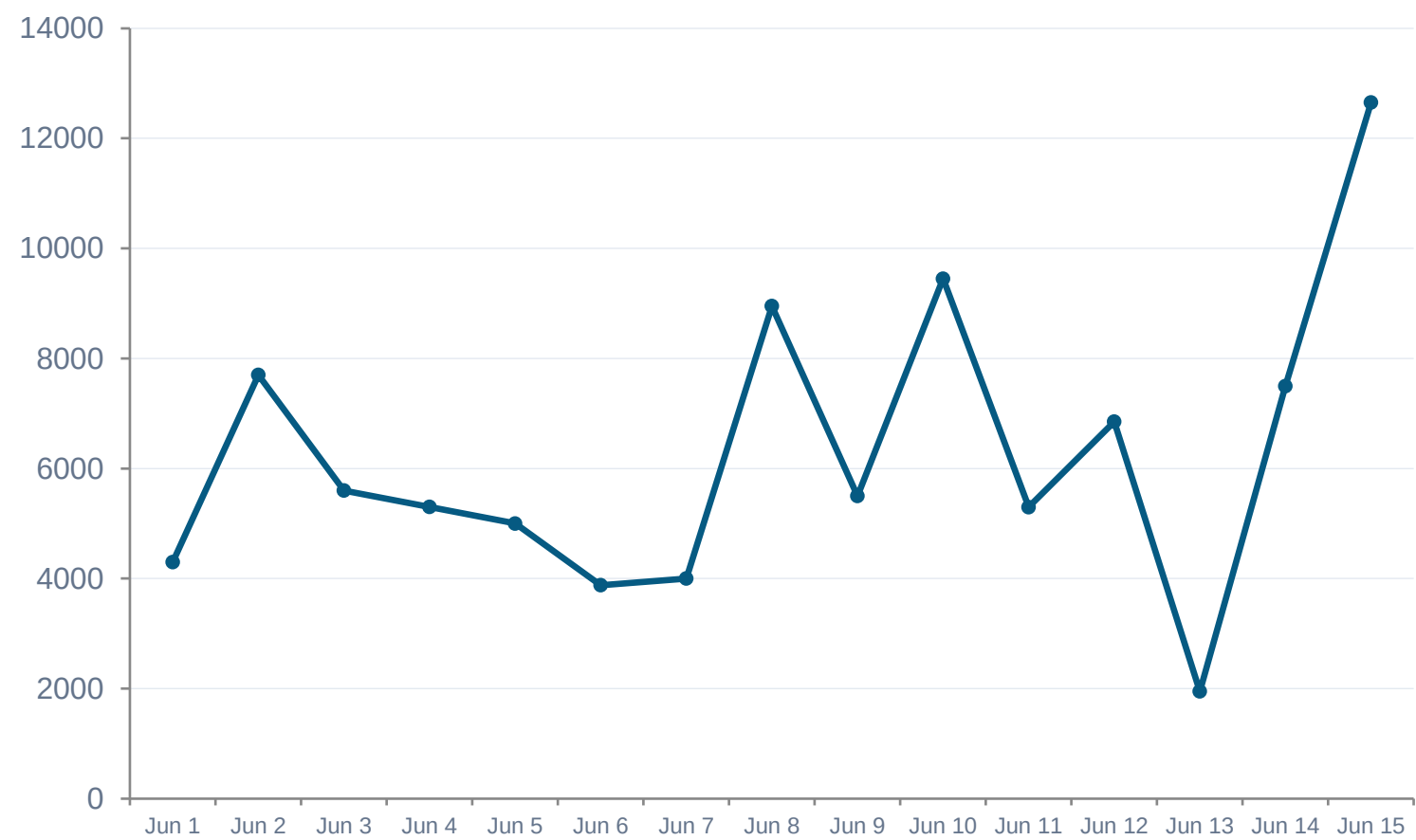
Lenovo ThinkPad rating drifted (4.8 → 4.7) between load batches — resolved by keeping the latest.



Type fixes

OrderDate / PaymentDate parsed to proper datetimes; text fields trimmed and case-normalized.

Daily Revenue Trend



Revenue counts only payments with Success status — Pending and Failed transactions are excluded.

₹93,933.53

Total Revenue (Success Only)

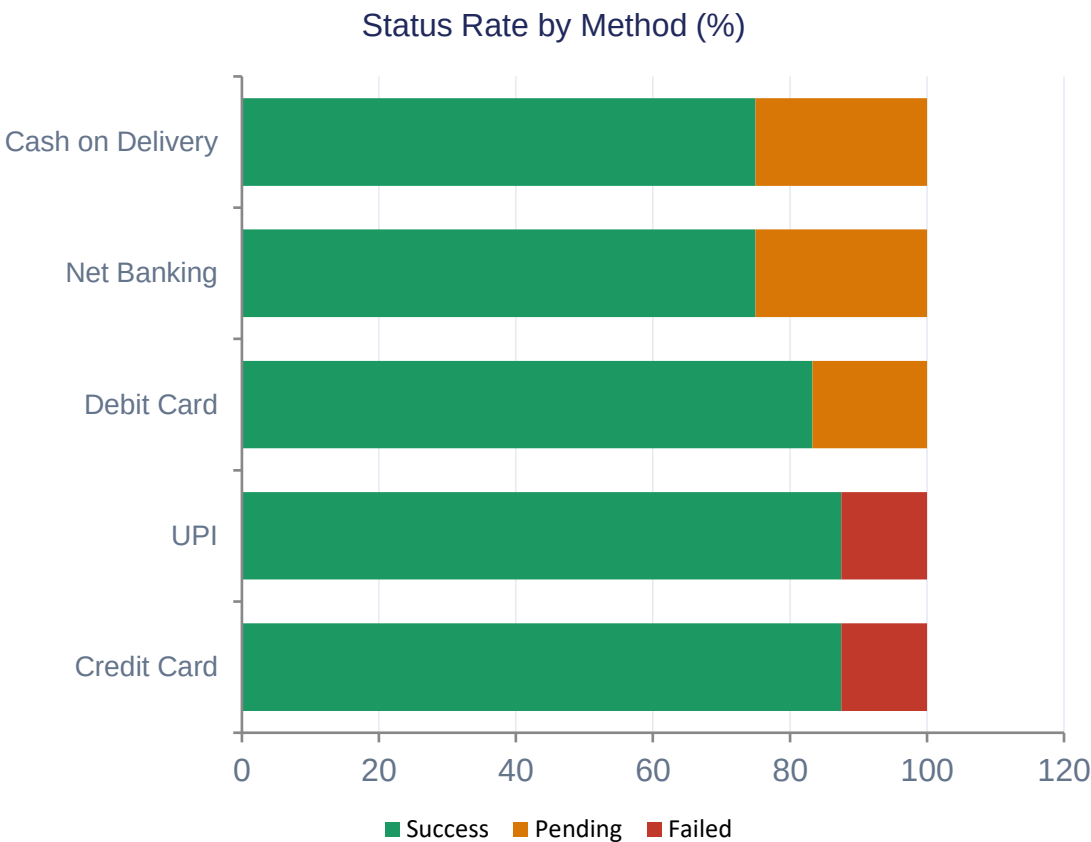
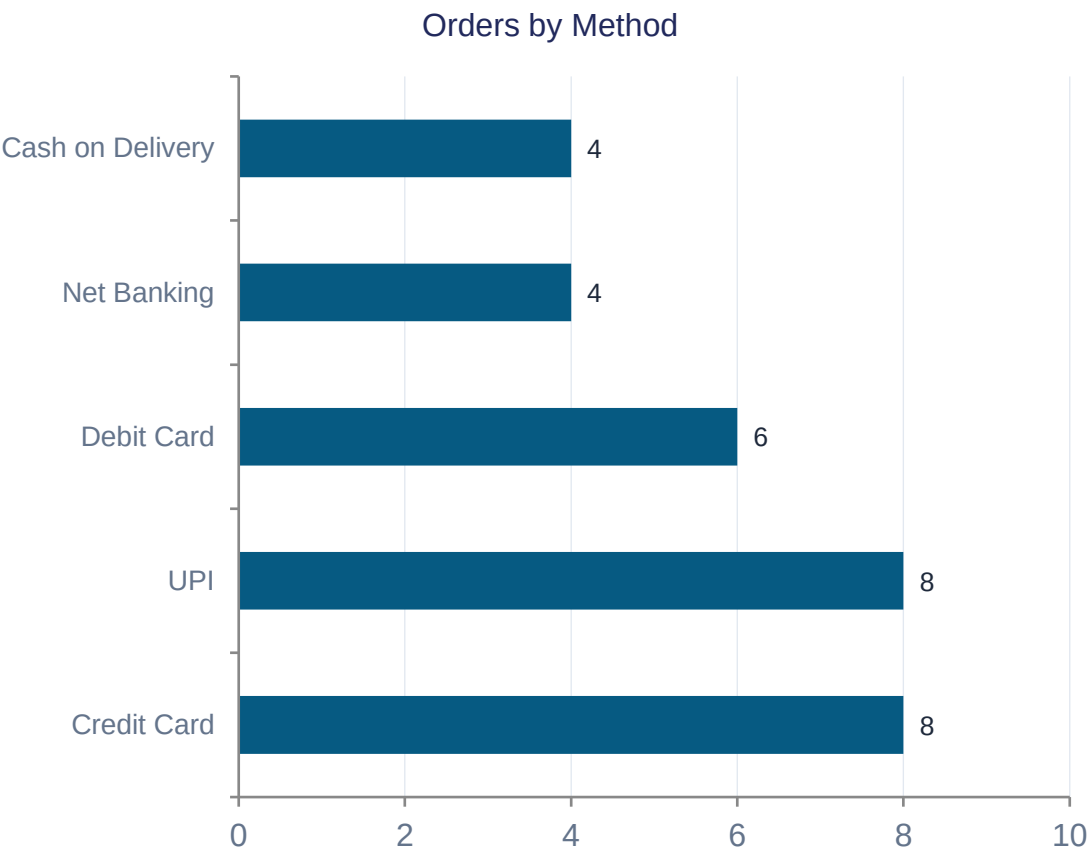
₹3,757.34

Average Order Value

Jun 15

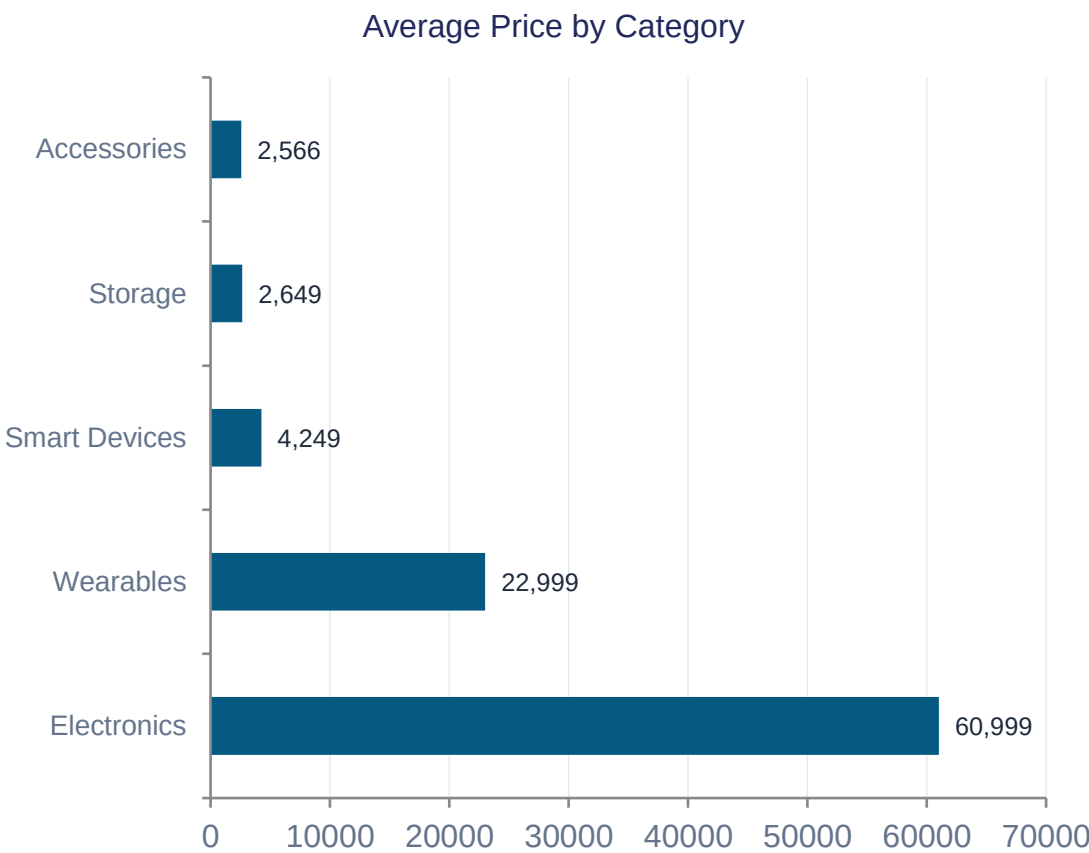
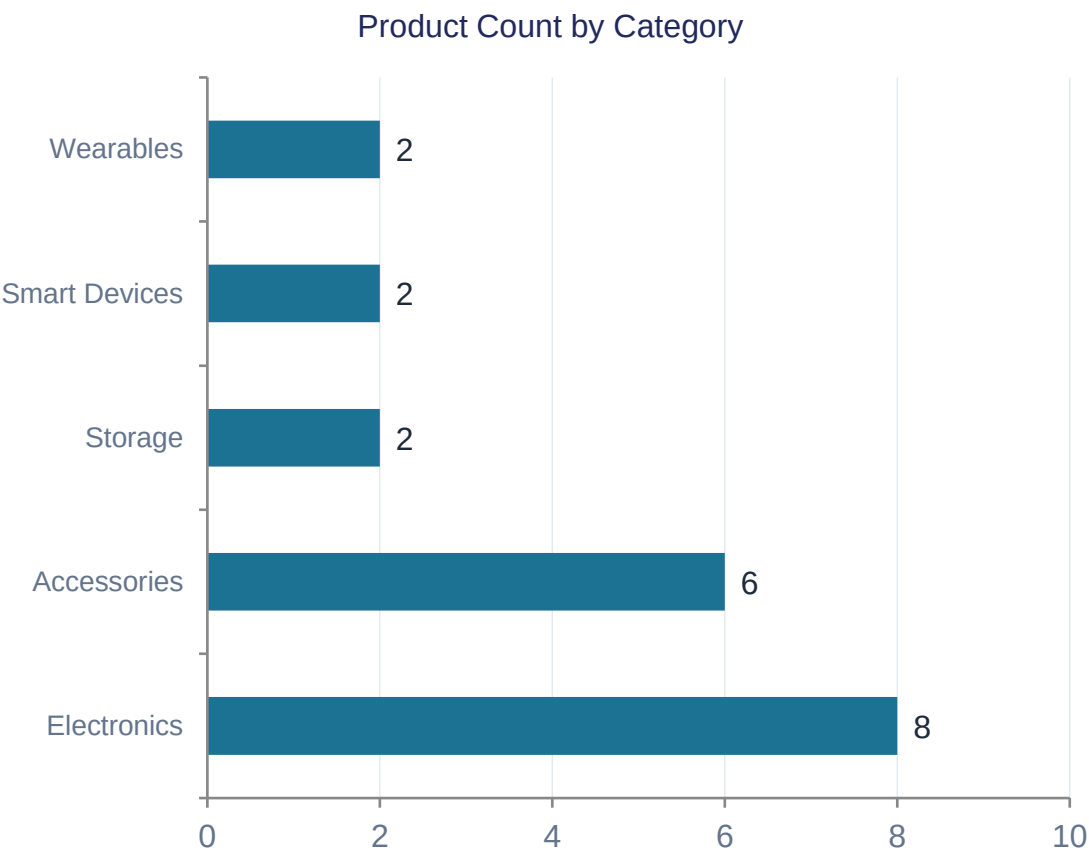
Peak Revenue Day (₹12,650.25)

Payment Method & Status Mix



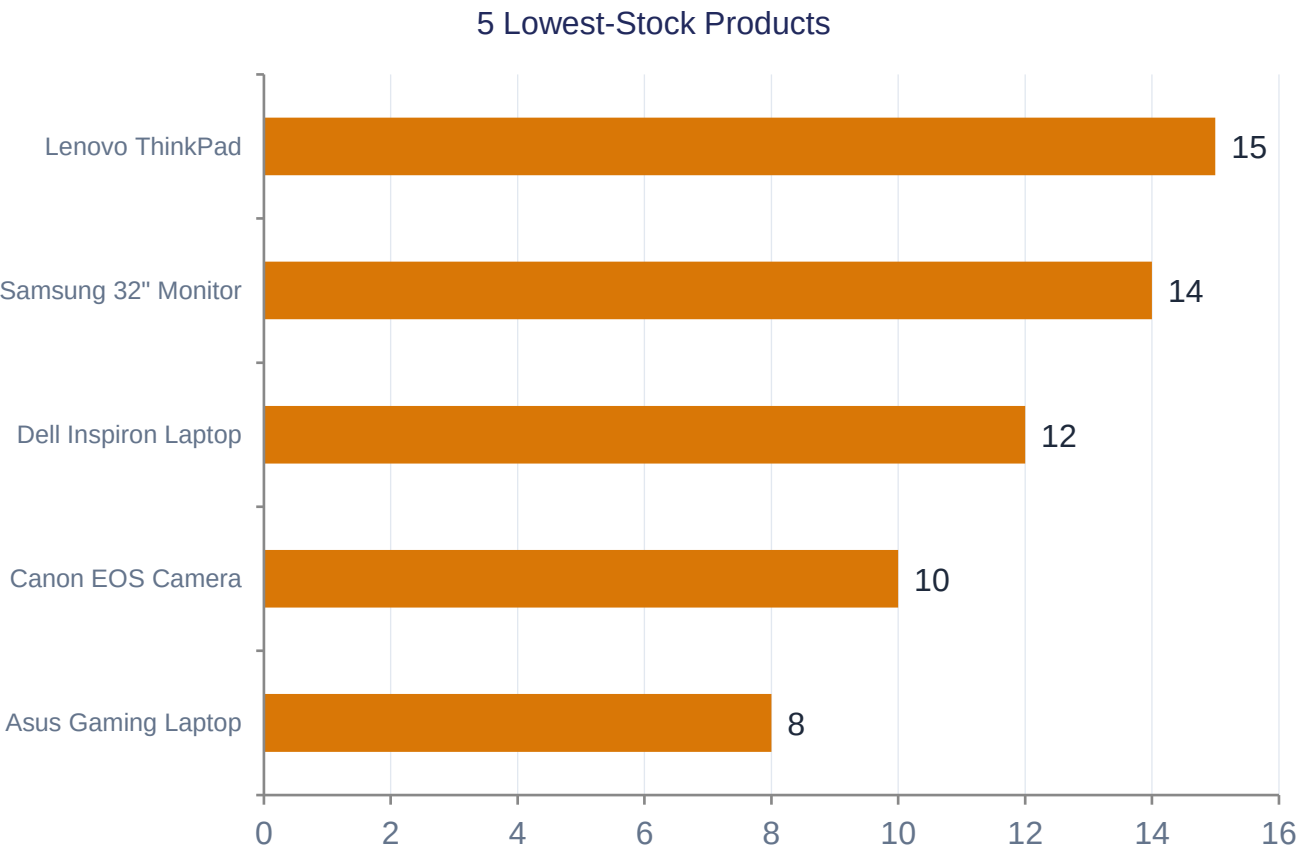
Credit Card and UPI lead in both volume and reliability (87.5% success); Net Banking and Cash on Delivery see the most Pending transactions.

Product Catalog by Category



Electronics dominates both inventory depth (8 of 20 SKUs) and average price point — nearly 3x the next highest category.

Stock Risk & Attribute Correlation



Price / Stock / Rating Correlation



Higher-priced products tend to carry lower stock and slightly higher ratings — consistent with premium, lower-volume electronics versus high-stock, lower-cost accessories.

Key Findings



Revenue grew 3x over the window

Daily revenue rose from ₹4,300 on Jun 1 to a peak of ₹12,650 on Jun 15.



83.3% of payments succeed

Credit Card and UPI are the most reliable methods; COD and Net Banking lag in completion rate.



Electronics drives the catalog

8 of 20 distinct products and the highest average price point (₹60,999) sit in Electronics.



5 products are at stocking risk

Asus Gaming Laptop (8 units) and Canon EOS Camera (10 units) are closest to stockout.

Tech Stack



SQL Server

Schema design, primary/foreign keys, relational integrity



Power BI

Data modeling, DAX measures, slicers, cross-page sync



Python (pandas)

Data cleaning, deduplication, merge & aggregation



Seaborn / Matplotlib

Trend lines, distributions, correlation heatmaps



Excel

Source data exports (Orders, Payments, Products)



Jupyter Notebook

Reproducible, executable end-to-end analysis

Thank You

Ecommerce Analytics — SQL Server, Power BI & Python in one pipeline

Deliverables: SQL schema • Power BI dashboard (.pbix) • Jupyter notebook (.ipynb)